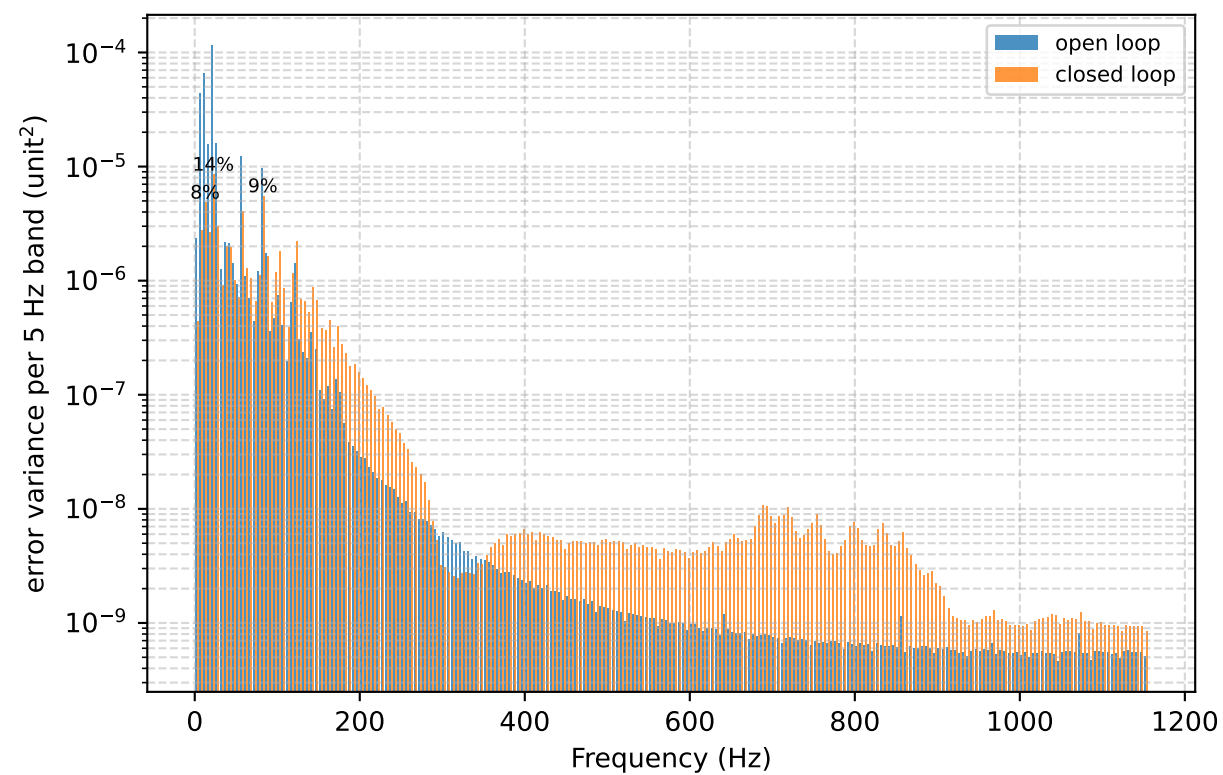
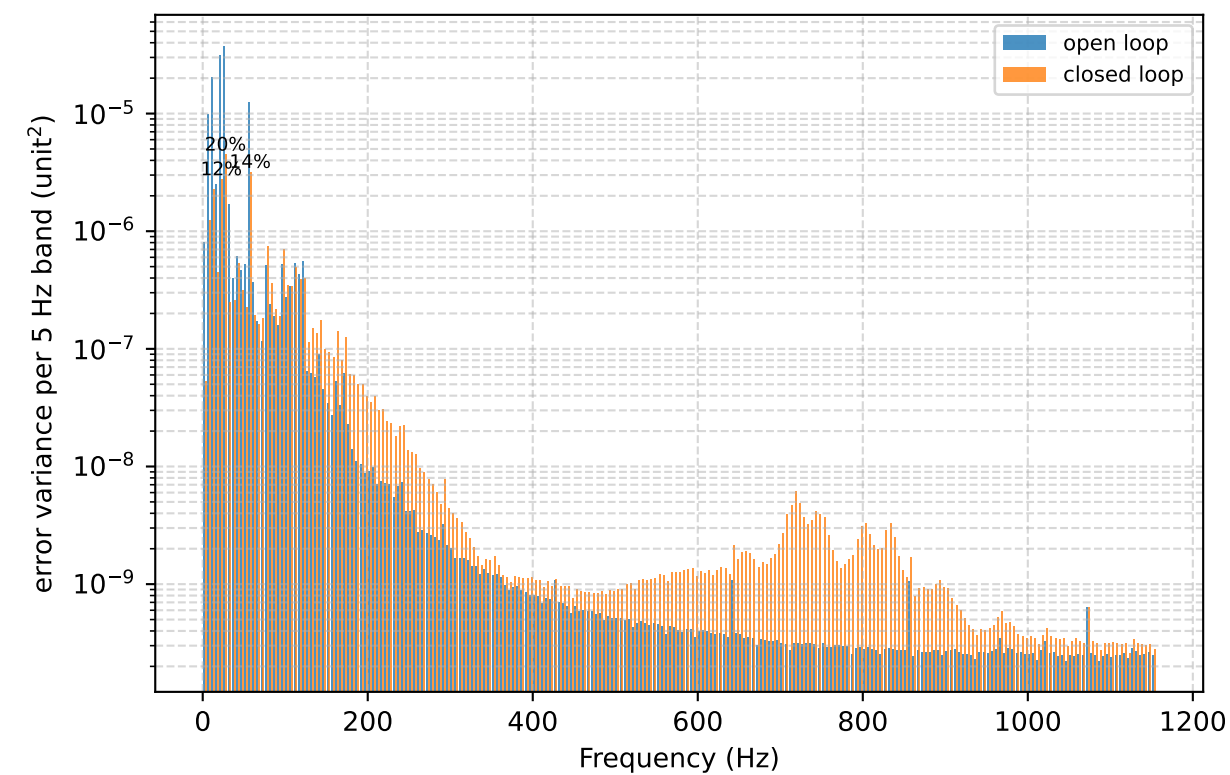
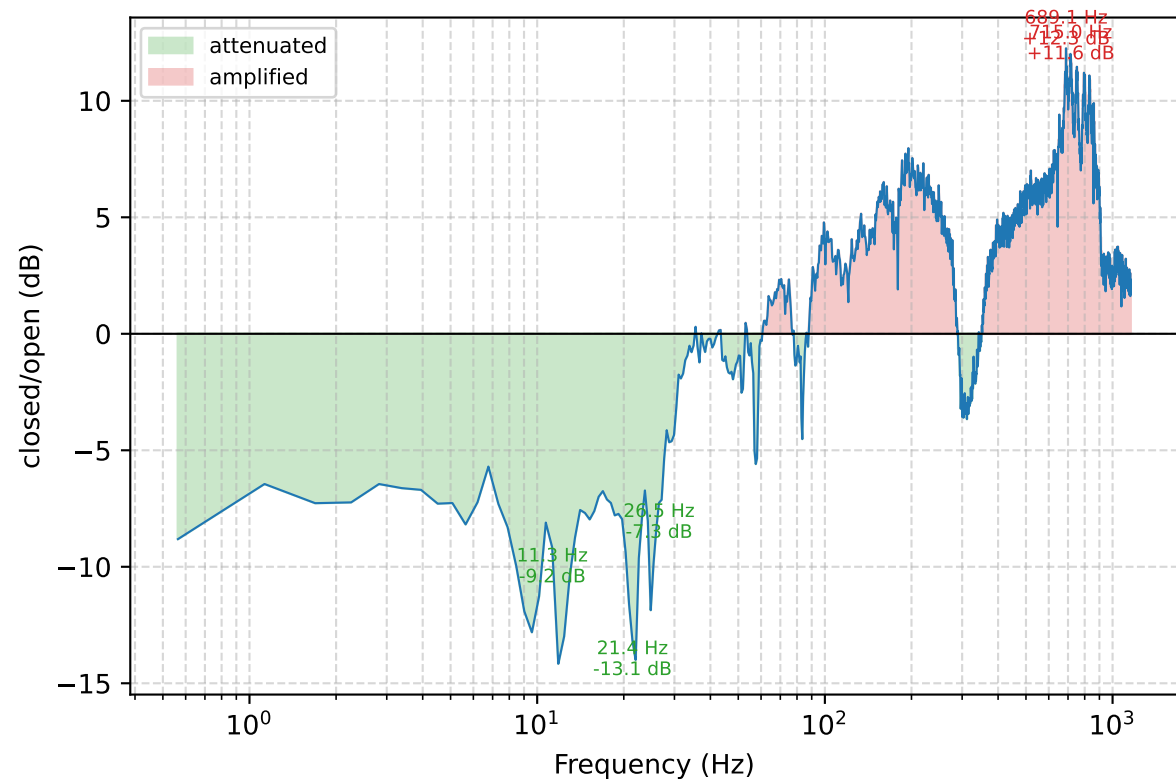
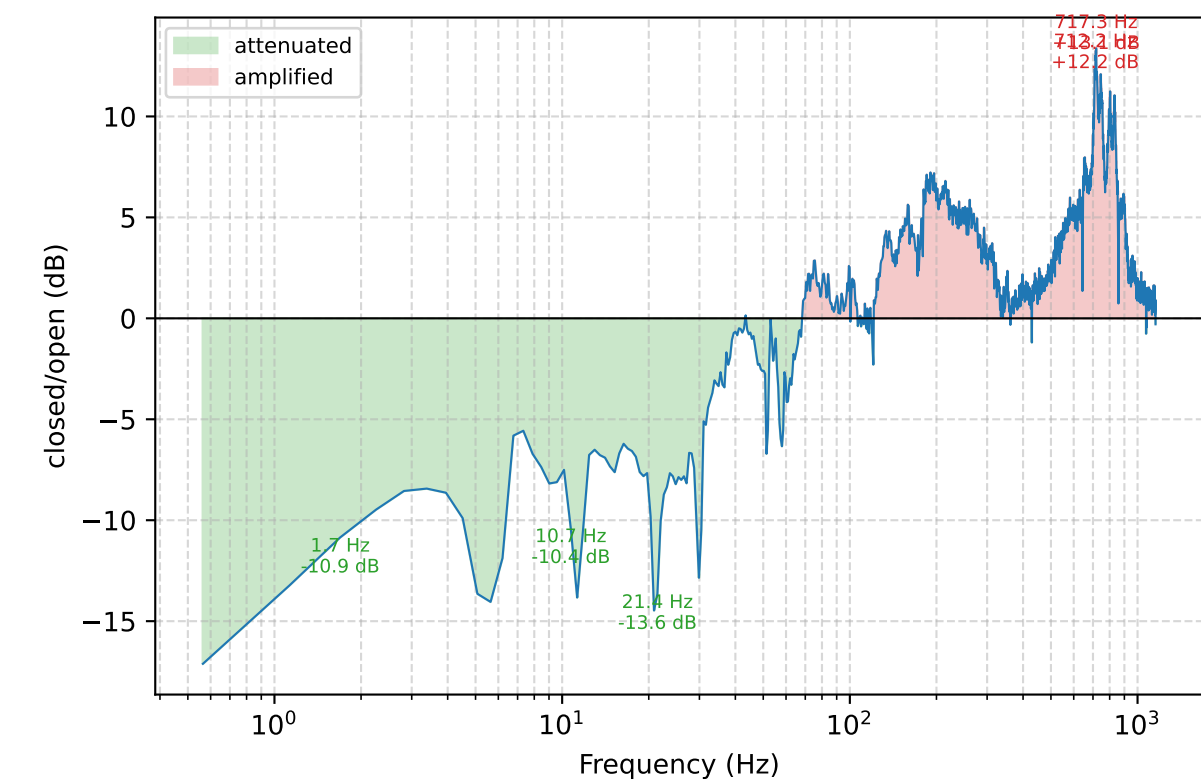
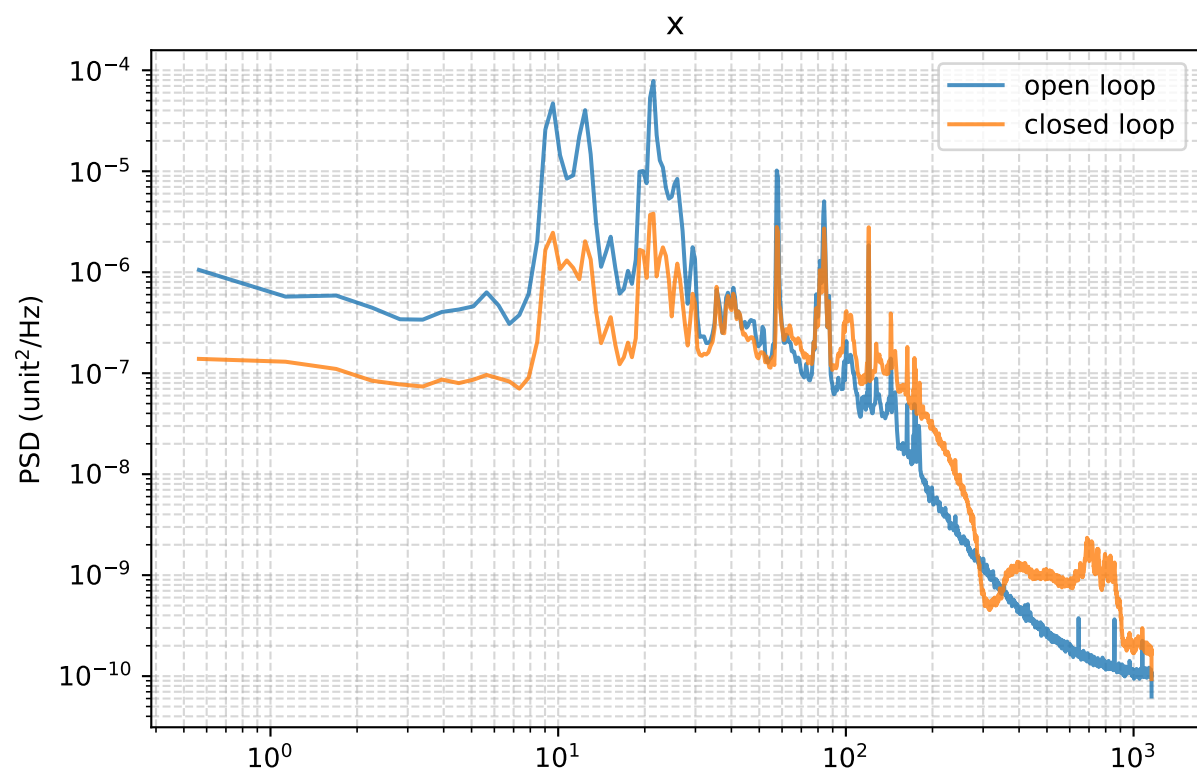
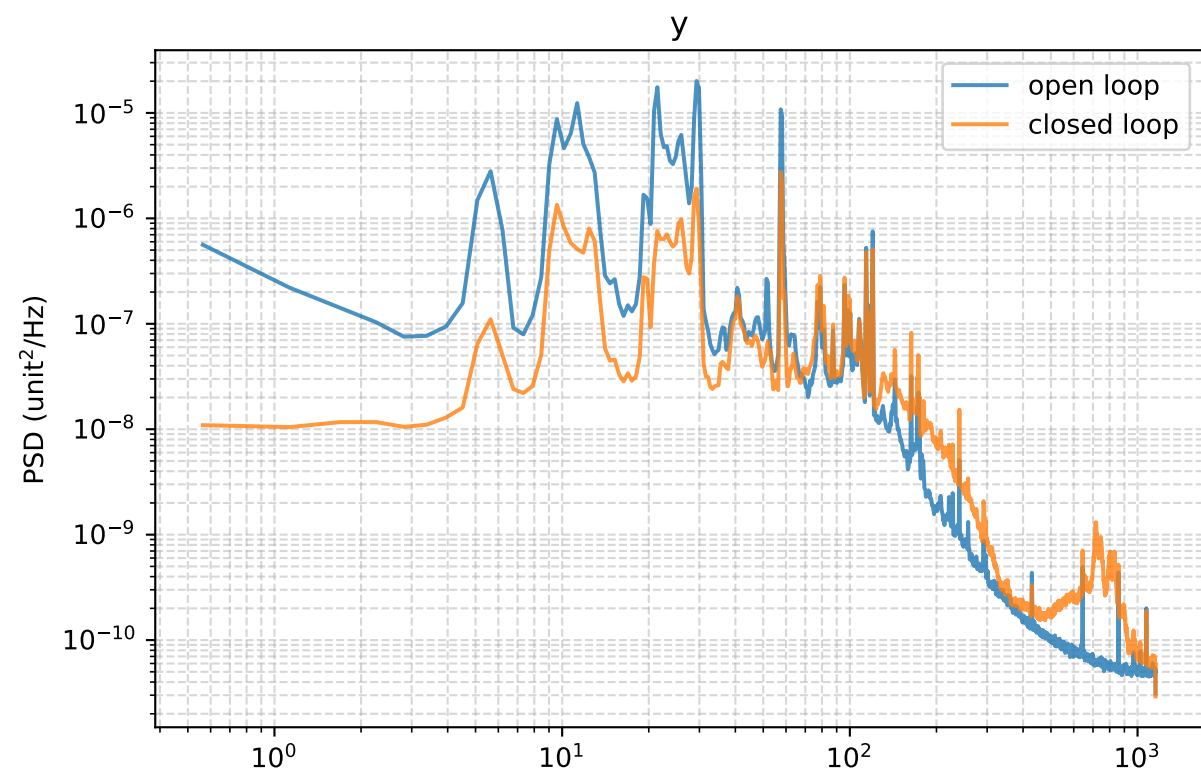


Closed-loop attenuation



Run configuration and per-band RMS

Started: 2026-07-14 12:46:09
pixel_scale: 31.5 px per output unit
y: mean +3.778 -> +0.000 px; RMS about mean 0.396 -> 0.151 px (open -> closed)
x: mean -7.494 -> +0.000 px; RMS about mean 0.566 -> 0.245 px (open -> closed)

Config used:
100, 500, 100, 500, 4096: camera ASI290MM 64x64, ROI start (920,488), ~2310 Hz loop
pid: i=1050 iy=1500 + LEAD (s+2pi*8)/(s+2pi*32) both axes -> fc 64/65 Hz, PM 67, model Ms 2.0
(design: grid search over attenuation9 open PSD; lead trades <8 Hz gain for +37 deg at 16-64 Hz)
lines x=[9.8,12.0,21.7,24.9,57.5,83.3] gain=[40,60,60,40,20,15] phase=[40.8,37.7,32.6,31.9,19.1,-2.7]
linesy=[5.64,11.3,21.1,29.6,57.5] gainy=[50,50,40,30,15] phasey=[52.3,38.6,32.8,31.0,19.3]
phases = plant+sensitivity WITH lead: phi=-360*f*tau + angle(1/(1+L))
kalman: order=80 horizon=5 mu=0.03 gain=1.0 gainy=1.0 ramp=10
piplate: fine ch0(+2)/ch1(-2), coarse bias 2.55/1.65 (bode @ 2.52/1.62), baud 460800

RMS contributed by each 10 Hz band (px)

band (Hz)	y open	y closed	x open	x closed
0-10	0.1032	0.0358	0.2140	0.0561
10-20	0.1512	0.0522	0.2827	0.0861
20-30	0.2619	0.0849	0.3604	0.1067
30-40	0.0456	0.0225	0.0582	0.0536
40-50	0.0327	0.0290	0.0593	0.0541
50-60	0.1138	0.0583	0.1146	0.0688
60-70	0.0232	0.0188	0.0421	0.0480
70-80	0.0251	0.0304	0.0406	0.0419
80-90	0.0207	0.0241	0.1057	0.0843
90-100	0.0262	0.0299	0.0287	0.0425
100-110	0.0247	0.0261	0.0336	0.0514
110-120	0.0310	0.0298	0.0288	0.0393
120-130	0.0248	0.0226	0.0413	0.0535
130-140	0.0109	0.0168	0.0210	0.0342
140-150	0.0116	0.0165	0.0244	0.0392
150-160	0.0078	0.0133	0.0141	0.0271
160-170	0.0092	0.0148	0.0138	0.0264
170-180	0.0092	0.0137	0.0155	0.0258
180-190	0.0050	0.0104	0.0097	0.0202
190-200	0.0044	0.0095	0.0082	0.0184
200-210	0.0043	0.0086	0.0074	0.0160
210-220	0.0038	0.0078	0.0066	0.0143
220-230	0.0038	0.0069	0.0060	0.0123
230-240	0.0035	0.0063	0.0056	0.0111
240-250	0.0034	0.0060	0.0052	0.0097
250-260	0.0029	0.0051	0.0048	0.0084
260-270	0.0024	0.0043	0.0043	0.0070
270-280	0.0023	0.0038	0.0040	0.0061
280-290	0.0022	0.0033	0.0038	0.0044
290-300	0.0023	0.0035	0.0035	0.0029
total 0-300	0.3511	0.1499	0.5448	0.2430